Ruminant Farm Systems (RuFaS)

Scientific Documentation

2026-04-01

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1 Preface: A Day in the Life on a RuFaS Farm

This document provides a high-level overview of the daily operations of a ruminant farm, as modeled by RuFaS. The overview will be through the eyes of Dr. Rue Fas. One day, Dr. Rue woke up and decided the life of a computer scientist wasn't for her. So she bought a dairy farm, and this is how she operates it every day.

Feeding the Animals Second thing in the morning, after coffee, Dr. Rue checks the calendar to see if it's time to formulate a new ration for her animals.

If a new ration needs to be formulated, Dr. Rue heads over to the barn where she keeps all her forages and feed. While there, she checks how well all the forages are holding up. After considering the feeds available, the quality of each one, and how much each one costs, she decides on how productive she wants her animals to be. It is a difficult balancing act, but eventually a ration recipe is found that will keep the animals happy while keeping costs low and sustainability high until the calendar says the ration recipe needs to be reformulated.

When Dr. Rue doesn't need to formulate a new ration, she takes the last ration recipe she made and grabs all the feed and forage the animals need for the day out of the feed barn. If there isn't enough for the animals, then a new ration is formulated using the steps above.

Taking Care of the Animals After making sure all the animals' feed is taken care of, Dr. Rue moves on to taking care of all the other animal tasks. This includes milking the animals, making sure they're healthy, getting them pregnant, and numerous other tasks.

As dawn breaks on the RuFaS farm, the animals awaken after a restful night. Joe, the farm master, prepares a large whiteboard to track HerdReproductionStatistics, ensuring he has all the data he needs for the day's activities.

Daily Routine - After leaving their pens, each animal lines up in front of the barn. Julie, the diligent farm veterinarian, examines them one by one. This daily check includes updates on the following areas:

- Nutrients: Each animal begins its day with a quick phosphorus requirement assessment. Julie records these details and updates the animal's phosphorus-related metrics.
- Digestive System: With nutrient information in hand, Julie shifts focus to each animal's digestive system, estimating manure output and enteric methane production for that day.

- Milk Production: All milking cows are assessed to gauge daily milk yield and milk component content (fat and protein). Any necessary adjustments are noted.
- Animal Growth: Around midday, Julie returns to each animal to evaluate daily weight changes. She updates each animal's body weight accordingly.
- Reproduction: HeiferII and Cow stage animals undergo a reproduction check. Julie closely monitors their reproductive progress and updates the HerdReproductionStatistics details on Joe's whiteboard.
- Life Stage Evaluation: Once the daily routines are complete, Julie determines if any animal is ready to progress to the next life stage. Graduating animals are marked as "graduated" to be moved to their new pens. Animals that must leave the herd—due to health, productivity, or other issues—are labeled as "sold" or "dead" to be removed from the farm population.
- Herd Size Maintenance: After every animal has been evaluated, Joe counts the herd to ensure it remains at a healthy size. If it is too large, he arranges the sale of some animals. If the herd is too small, he may opt to purchase additional heiferIIIs to keep numbers at the ideal level.
- Herd Structure Management: Joe also manages pen assignments. Newly graduated or newborn animals are guided to suitable pens, while animals marked for removal (sold or deceased) are taken out of the herd and removed from the farm records.

Output Reporting At day's end, Joe and Julie review the updated HerdReproductionStatistics on the whiteboard. Joe then compiles a concise report summarizing the day's herd stats. He forwards this information to the OutputManager before wrapping up another successful day on the RuFaS farm.

Dealing with the Manure Once all the animal chores are finished, Dr. Rue moves on to handling the manure. There are a lot of different ways that manure is collected from the animals: it is flushed out of the milking parlor with water, some of the pens need to have it shoveled out along with the bedding in the pen, and some pens have grates that allow the manure to go straight into a pit beneath the barn.

There are a lot of different places where manure is processed and stored on the farm. Dr. Rue goes from place to place on the farm, moving the manure from one place to the next. The manure chores are finished after moving all of the manure into her lagoons, compost piles, and other storages. As this final manure chore is completed, she writes down all the information about how much manure is in all these storages, and some information about what that manure is made of. How much water, nitrogen, phosphorus, other solids, etc. are in each storage.

Managing the Fields Now that Dr. Rue is done making sure all the manure has been handled and stored properly, she can take care of all the field tasks. She has a lot of fields on her farm, each with its own unique schedule. Her field management routine starts with checking each field's schedule to see if any of them are due to have manure applied, and if so, how much they should have applied. After noting all the manure that is supposed to go onto her fields, these amounts are checked against the amounts of manure that are in her manure storage. Dr. Rue then applies manure to each field on the schedule that day, trying to match the amount applied to the amount on the schedule as best she can.

After making sure every manure item on each farm's schedule has been taken care of for the day, the schedule for each field is rechecked, one field at a time. In each schedule she checks to see if that field needs to be tilled, if any chemical fertilizer needs to be applied, and if any crops need to be planted or harvested. After all these activities are taken care of, Mother Nature goes to work. The sun shines and rain falls, providing the field with water and energy that grows crops and soaks the soil. After taking care of each field one by one, Dr. Rue drives back to the main farm with all of the crops that she harvested on that day.

Storing the Feeds Now that she is back at the barns, all the crops that were just harvested can be stored. Fortunately she is very organized, so all her harvested crops are separate from one another and each has a label, indicating where it will be stored inside the feed barn. As each feed is stored, it is labeled with the current date. Dr. Rue is exhausted from another long day. She goes back to the farmhouse and rests up, preparing to do the whole thing over again tomorrow. Her daily routine might seem repetitive, but each day is a crucial part of the larger, continuous process that keeps her farm running efficiently and sustainably.

2 New Version Updates

Each version of the RuFaS model is designed to improve the accuracy and precision of the predictive power of the model. This section aims to outline the changes made …

2.1 Animal Module

2.1.1 Changed Inputs

Here you should include any

2.1.2 Changed Methodology

Text, equations, and references

2.1.3 Output Updates

Expectations

2.2 Manure Module

2.2.1 Changed Inputs

Here you should include any

2.2.2 Changed Methodology

Text, equations, and references

2.2.3 Output Updates

Expectations

2.3 Feed Storage Module

2.3.1 Changed Inputs

Here you should include any

2.3.2 Changed Methodology

Text, equations, and references

2.3.3 Output Updates

Expectations

2.4 Crop and Soil Module

2.4.1 Changed Inputs

Here you should include any

2.4.2 Changed Methodology

Text, equations, and references

2.4.3 Output Updates

Expectations

2.5 Connections Documents

2.5.1 Changed Inputs

Here you should include any

2.5.2 Changed Methodology

Text, equations, and references

2.5.3 Output Updates

Expectations

3 Example Inserts

3.1 Sectioning and Structure

There is some variation between modules and documentation bassed on structure and needs of the model. However there are some materials, or ingredients if you will, that will remain consistent between modules. These include: Module Header, module introduction, sections representing components of module, section introductions, methodologies, relevant inputs, and relevant outputs.

These should be referred to in a consistent manner throughout the document to inform tables of contents, indices, and auditing materials.

```
# Module Name --> section title and only appears once

## Module Introduciton --> subsection header

## Each Section Title --> subsection header that introduces distinct portions of each module th

### Section Introduction --> subsubsection header to introduce the concept

### Methodology --> subsubsection header used to start descriptions

**Relevant Inputs or Relevant Outputs** --> emphasized headers should be used to report the inp
- Variable
- Definition
- Default
- Min
- Max

If there is no application for one of these (default, min, or max particularly), do not leave t

*Alternate Headers* alternate headers may be used at the discretion of the author
```

There may be expanded uses for subsubsection headers and emphasized headers. The ones outlined above are only the expected uses. Feel free to use more at the discretion of the author.

3.2 Images, Figures, and Tables

3.2.1 Example of a Figure

All figures and images must be uploaded in the “references.” When you upload your fig/img, you may name the file (best to keep it short and sweet)…

```
![Animal module icon](resources/animal.png){#fig-animal-icon width=35%}

-->[Caption for the image or figure]
--> #fig-animal-icon = #indicates the type of file-label-for-auditing
--> width=35% = sizing adjusted as needed
```

The output is below :

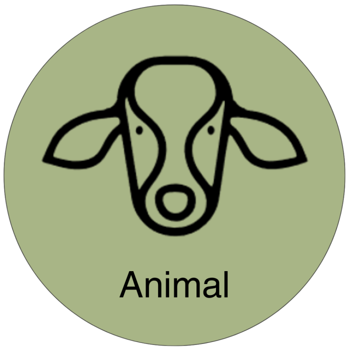

Figure 3.1: Animal module icon

3.2.2 Example of a Table

```
| Variable | Definition |
|---|---|
| heiferI_num | Number of Heifer I animals |

: Example animal inputs {#tbl-animal-inputs}


--> The top row of text is your headers and will appear centered and bolded unless otherwise co
--> The  next row of actual text is what appears in the row with the associated column. Quarto
--> Be sure to include ..... for between each row

--> ": Example animal inputs" will appear as the caption. Be as descriptive as possible so tha
--> #tbl-animal-inputs = #indicates the type of file-label-for-auditing. Though you may label i
```

The above markdown text will appear as it does below:

Table 3.1: Example animal inputs

Variable	Definition
heiferI_num	Number of Heifer I animals

3.3 Equation Templates and ID Logic

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.2)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.6)}$$

See Equation 5.1.

3.4 Callouts, Recommendations, or Notes

You may want to emphasize specific notes and annotations. Here is a helpful tool to do so correctly.

```
::: {.callout-note}
Callout text goes here
:::
```

Your callout will appear the way it does below:

Note

Note: if only interested in a PDF version of the documentation (no website version), then the original LaTeX code can still be used and processed by quarto. For example, the bullet points works as the original LaTeX or translated into markdown syntax:

````
```{latex}
\begin{itemize}
    \item first bullet point
    \item second bullet point
    \item ...
\end{itemize}
```
````

````
```{markdown}
* first bullet point
* second bullet point
* ...
```
````

- first bullet point
- second bullet point
- …

but only the markdown works for all document types (html, pdf, word document, presentations, e-books, etc.).

3.5 References

Before adding a new reference to "rufas.bib," check the file to ensure it has not already been added. All references should be listed alphabetically and then by year (most recent at the top).

If you do need to add a bibliography, the correct structure is below. Pay special attention to the green text, this formatting is important to maintain consistentcy because it helps us with our auditing report and allows us to properly cite in-text.

```
Journal Articles
@article{LASTNAME_YEAR,
  author = {LASTNAME, FIRSTNAME},
  title = {Title of the article in proper formatting},
  journal = {Journal Title with proper NCBI Abbreviations},
  year = {YEAR PUBLISHED},
  volume = {VOLUME NUMBER},
  pages = {PAGES}
}

Books

@book{LASTNAME_YEAR,
  author = {FIRSTNAME LASTNAME},
  title = {Title of Book Referenced},
  publisher = {Publisher},
  year = {YEAR PUBLISHED},
  isbn = {978-3-319-24277-4}
}

Online, Misc, Website (RARE USAGE)

@online{quarto2024,
```

```
  author = {Posit},
  title = {Quarto Citations Guide},
  year = {2024},
  url = {https://quarto.org/docs/authoring/citations.html}
}
```

If there are 2 articles with the same author and same year, add the recommended “a” or “b” (e.g. `bibtex @LASTNAME_YEARa vs @LASTNAME_YEARb`) as they are referenced.

4 Equation examples (from Animal Documentation)

4.1 Nutrient Requirements of Dairy Cattle (NRC)

Energy Requirements For details about the variables included in each calculation, please refer to the energykey.

- ***Maintenance*** - The maintenance requirement is calculated based on metabolic body weight (body weight$^{0.75}$).

  Lactating and Dry Cows

$$CBW = MW * 0.06275$$

$$CW = (18 + (DOP - 190) * 0.665) * \left(\frac{CBW}{45}\right), \text{if } DOP > 190.0$$

  Otherwise

$$NEmaint = 0.08 \times (BW - CW)^{0.75}$$

4.2 Reformatted equations

Equation 4.1 may be visually improved by using the `\times` LaTeX command:

$$CBW = MW \times 0.06275$$

4.3 Use alphanumeric equation labels

Though Quarto does not allow custom cross-reference labels, you can include your own non-CR label with the `\tag` LaTeX command. The equation can still be cross-referenceable using a standard Quarto equation ID.

$$CBW = MW * 0.06275$$

See Equation AN.NRC.2.

4.4 Other equation Templates

The energy balance is $E = mc^2$.

$$E = mc^2$$

See Equation Equation 4.5.

$$\text{nutrient\_proportion\_to\_be\_removed} = \text{min}\left(\frac{\text{mass\_requested\_nutrient}}{\text{manure\_nutrient\_available}}, 1\right)$$

5 Animal Module

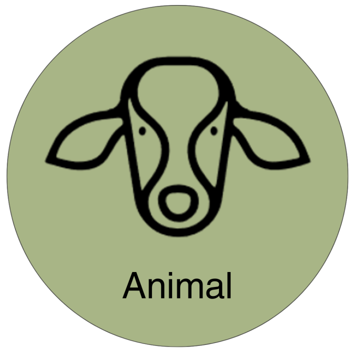

Figure 5.1: Animal module icon

5.1 Introduction

The Animal Module simulates individual animals on a daily basis throughout each stage of life from birth until removal from the herd. In addition to the dynamic Monte-Carlo models that determine the occurrence of events throughout each animal's life, the Animal Module uses process-based, dynamic models to simulate the following processes:

- Individual animal growth
- Gestation and milk production
- Estimation of nutrient requirements and feed intake
- Enteric emission and manure excretion

5.1.1 Animal Life Cycle

The animal life cycle is driven by Monte Carlo stochastic processes that simulate the growth, production, reproduction, and culling of individual dairy cattle animals and shares aggregated outcomes with the herd management sub-module to manage herd dynamics on a daily basis.

The life cycle module represents the variation in animal performance through the use of Monte Carlo Simulation methods [Reuven and Kroese (2017)}. Monte Carlo simulation relies on random draws from probability distributions rather than assigning fixed values. In this model, two main strategies are used:

- Event-based, in which a random draw from a uniform distribution (U(0,1)) is compared to the probability of an event occurring to simulate if that event occurs, and

- Population-based, in which a random draw from a known distribution of attribute values is assigned to an animal at instantiation. Probability distributions used for that purpose are Normal Gaussian or Empirical.

Each animal within the RuFaS herd progresses through five main life stages (calf, Heifer I, Heifer II, Heifer III, and Cow) as illustrated in the figure below.

- Calves progress to Heifer I when their age reaches the user-defined input for weaning day
- Heifer I progresses to Heifer II when their age reaches the user-defined input for breeding start day
- Heifer II animals that successfully conceive become Heifer III's when the number of days left in their pregnancy is equal to the user-defined input for prefresh days (default 21)
- When a Heifer III calves, she becomes a Cow for the rest of her life on the RuFaS farm
- Cows cycle through lactating and non-lactating phases until they are removed from the herd due to a failure to conceive or a user-provided probability for sale or death within each parity (lactation).

When a HeiferIII or Cow reaches full gestation length, they start lactating and a new calf object is created. The calf is assigned a birthweight according to a user-informed random distribution. All male calves are recorded and removed from the farm on day 1 of their life. Female calves either remain on the farm or are sold on day 1, determined by an event-based Monte Carlo method informed by the user-defined percent of female calves that are kept.

5.2 Herd-level Processes and Management

5.3 Herd Initialization

> **Note**
>
> In RuFaS code, the term "herd initialization" is used to refer to both the process of generating an animal population file and of selecting from that population to create a specific herd at the start of a simulation. In this doc, those two processes will be differentiated by calling them "Generation" and "Selection", respectively.
> "Animal population" can refer to both groups of animals (the pool of animals generated, and the herd of animals selected). In the code, the population is sometimes referred to as "pre_animal_population" and "post_animal_population" to differentiate whether the specified group of animals exists before or after the random selection of individuals to create a herd.

5.3.1 Introduction to Herd Initialization

Every simulation begins with a herd consisting of calves, heifers, and dry and lactating cows. This herd comes to be through a two-step process:

- A population of animals is generated through simulation or loaded from data (pre-animal-population). The animals in that herd should reflect the attributes of animals in the herd to be modeled.
- Animals are randomly selected with replacement from the pre-animal-population to be in the initial herd, based on group numbers and characteristics that reflect the distribution of animals within the herd being modeled. Note that the animals in the herd on day one will have attributes of the pre-animal-population and may deviate from the desired herd to be modeled.

For the rest of the simulation, animals enter the herd through birth as newborn calves or purchased replacements (heiferIII's).

Table 5.1: Overview of the distinct phases by which animals are created and introduced to the herd.

Before Simulation	Simulation Initiation	During the Simulation
Generate (simulate) or load an Animal Population file (pre-animal-population). The animals present at the start of the simulation, and those available in the replacement market, are drawn from this file.	Herd is established by random selection of animals from the Animal Population to match the user-specified group sizes and age/stage distribution (post-animal-population). Management attributes, including lactation- curve parameters, are assigned as each animal enters the herd.	New animals enter via (a) birth or (b) purchase| of heifers from the | replacement market. Purchased animals inherit| two attributes from the | population file, | - Breed | - Mature bodyweight |

Relevant Inputs

Table 5.2: User inputs in the animal input file that influence the generation of the Animal Population (pre-animal-population). The inputs of this table are found in the Herd Initialization Section.

Variable	Definition	Default	Min	Max
Initial_animal_num	The initial number of animals that serve as a starting point for the population generation simulation	10000	0	-
Simulation_days	The number of days to simulate for the population generation simulation	5000	0	-

Variable	Definition	Default	Min	Max
Calf_num	Number of Calves (head) – The initial number of pre-weaned calves	8	2	–
HeiferI_num	Number of HeiferI animals (head) – The initial number of heifers that are weaned but not yet bred	44	2	–
HeiferII_num	Number of HeiferII animals (head) – The initial number of heifers that are either eligible for breeding (based on user-inputted Breeding Start Day for heifers), or in early pregnancy	38	2	–
HeiferIII_num_springers	Number of HeiferIII animals (head) – The initial number of close-up heifers (within the user-defined close-up period, e.g., Prefresh Day)	5	2	–
Cow_num	Number of Cows (head) – The initial number of dry and lactating cows	100	6	–

Variable	Definition	Default	Min	Max
Replace_num	replacements (head) – Number of replacement animals available in the replacement market	5000	50	–
Lactating_fraction	Fraction of cows that are lactating. Used during herd initialization	0.8356	0	1
Parity_fractions: 1, 2, 3, 4, 5	Fractions of the milking animal population that are parity 1, 2, 3, 4, and 5+. The sum must equal 1.0	0.36, 0.26, 0.18, 0.11, 0.09	0	1

Relevant Outputs

After the animals have been selected from the animal population to create the initial herd, a summary of the Population (pre-selection pool, aka pre-animal-population) and of the Initial herd (selected animals, aka post-animal-population) is provided to the output manager.

Class and function: AnimalModuleReporter.report_animal_population_statistics

Output variables: - .population_{…} - .initial_{…}

{…} includes: breed, number of animals of each class (calf, heiferI, heiferII, heiferIII, cow), number of cows by parity and lactation status, average age of animals within each class, distribution of age of animals within each class, average body weight of animals within each class, average cow days in milk, days in preg, parity, and calving interval.

5.3.2 Methodology

When a simulation begins, the HerdFactory class generates or loads the pre-animal-population, from which it selects and compiles the post-animal-population with an appropriate distribution of animal ages and stages to serve as the starting herd for simulation.

Designation of an Animal Population

- Option 1: Creating a new Animal Population An animal population can be generated as a stand-alone simulation task that creates and saves a pre-animal-population file as the output based on the user Animal Module inputs. Or, the animal population can be generated as the first step of a regular simulation task which then uses that generated pre-animal-population to select from and create the post-animal-population to use for the subsequent herd simulation.

See the documentation about Task Manager for more information about how to generate a new pre-animal-population.

The _generate_animals() method creates the pre_animal_population, with the option to save those animals to file at the end of the generation process.

Calves are created according to the number specified by the user (initial_animal_num), some are sold based on user inputs, and those remaining proceed through select daily updates (growth, reproduction, milk, and transition through the appropriate life stages) for the user-specified number of days (simulation_days). The simulation occurs similarly to the regular life cycle simulation process, but with an extended period and large number of animals. There are a few notable distinctions:

```
- After 3000 days of simulation (specified in animal_constants), a copy of each heiferIII is sa
- If a cow has made it to parity six without being culled, she is removed from the herd. (The r
```

After the simulation_days number of days, the full herd contains a mixture of calves, heifers, cows, and replacements. The core status information and life history of each animal are either saved to file or passed for random sampling to create the post-animal-population.

- Option 2: Point to an existing Animal Population Alternatively, the pre-animal-population will be loaded from a specified Animal Population input file consisting of calf, heifer, cow, and replacement objects.

Herd Selection The _random_sample_with_replacement() method selects animals from each of the following groups: calf, heiferI, heiferII, heiferIII, replacement, lactating cows by parity for parities 1-5, and dry cows by parity for parities 1-5.

Animals are sampled randomly with replacement from the pre-animal-population according to the numbers, parity fractions, and milking cow fraction specified in the animal input file. The resulting list of animals is returned as the post-animal-population and summaries of the pre- and post-animal-populations are reported to the output manager.

5.4 Pens and Animal Grouping

5.4.1 Introduction to Grouping

All animals in RuFaS are assigned to a pen which is used to summarize nutrient requirements for diet formulation and feeding, aggregate enteric methane and manure excretions for reporting and passage to the Manure Module. Each pen tracks and updates daily a list of animals in the pen, the stocking density, the average nutrient requirement of the animals in the pen, the ration being fed to that pen, the manure excreted, and the methane emitted. In addition, constant attributes of each pen include the type of pen it is (freestall, tiestall, open lot, compost bedded pack barn), the type of animals that can be in that pen, the number stalls, the manure management practices associated with that pen, and the distance to the milking parlor for lactating cows. Each day after the HerdManager has updated the status of each animal and determined which animals need to be bought or sold, the animals are added and removed from pens as needed.

Relevant Inputs

Variable	Definition	Default	Min	Max
id	The index of the pen	NA	0	NA

pen_name | The name or identifier of a given pen | NA | | |

animal_combination | The valid combinations of animal types that can be in a pen. CALF = Calves; GROWING = HeiferI's and HeiferII's; CLOSE_UP = HeiferIII's and Dry Cows; LAC_COWS = Lactating Cows | | CALF, GROWING CLOSE_UP, LAC_COW |

vertical_dist_to_milking_parlor | The elevation gain (in meters) between the pen and the milking parlor | 0.1 | 0 | NA |

horizontal_dist_to_milking_parlor | The distance (in meters) between the pen and the milking parlor | 10 | 0 | NA |

number_of_stalls | The number of stalls in the pen | 1000 | 0 | NA |

pen_type | The type of pen (freestall, tiestall, open lot, compost bedded pack barn) | freestall | freestall, tiestall, open lot, compost bedded pack barn |

max_stocking_density | The maximum ratio of the number of animals in the pen to the number of stalls in the pen | 1.2 | 0.5 | 5 |

manure_management_scenario_id | The ID number of the manure management practices that are used for this pen (scenarios are defined in manure management schema) | 0 | 0 | NA |

: User inputs in the animal input file that influence pen grouping. The user must define at least 3 pens and for each pen, key inputs are outlined here. {#tbl-animal-grp-inputs}

Relevant Outputs

- number_of_animals_in_pen_[id]_[AnimalCombination] reports the number of animals in each pen each day.
- ration_per_animals_for_pen_[id]_[AnimalCombination] - reports average dry matter intake of an animal in the pen at the time of the ration formulation.
- ration_per_animals_for_pen_[id]_[AnimalCombination].[RUFAS_feed_id] - reports the amount of each feed that is delivered per animal in the pen
- ration_nutrient_amount_for_pen__[AnimalCombination].[nutrient] - is reporting the amount of each of the following nutrients that is provided by the ration per animal in the pen (all reported in kilograms unless otherwise indicated)

5.4.2 Methodology

5.4.3 Examples of Equations

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.2)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.3)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.4)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.5)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.1)}$$

$$CBW = MW * 0.06275 \qquad \text{(AN.NRC.6)}$$

See Equation 5.1.

6 Manure Module

6.1 Introduction

6.1.1 Example of a Figure

6.1.2 Example of a Table

Table 6.1: Example animal inputs

Variable	Definition
heiferI_num	Number of Heifer I animals

Reuven, Y. R., and D. P. Kroese. 2017. *Simulation and the Monte Carlo Method. In: Wiley Series in Probability and Statistics*. John Wiley; Sons Inc.